Data Science Project

$$P = \frac{1}{(1+e^{-z}) \Rightarrow u} = \frac{1}{u} = u^{-1}$$

$$\boxed{P = u^{-1}}$$

Splitting into two layers:

① Outer function:- $\frac{dp}{du} = \frac{d}{du}(u^{-1})$

$= -1u^{-2}$

$= -1(1+e^{-z})^{-2}$

② Inner function $= \frac{d}{dz}(1+e^{-z}) = 0 + (-1)e^{-z}$

$= -e^{-z}$

Now, $\frac{dp}{dz} = \frac{dp}{du} \times \frac{du}{dz}$

$= -1(1+e^{-z})^{-2} \cdot -e^{-z}$

$= \frac{e^{-z}}{(1+e^{-z})^2}$

$= \frac{1}{(1+e^{-z})} \cdot \frac{e^{-z}}{(1+e^{-z})}$

$= p \cdot \frac{e^{-z}}{1+e^{-z}}$

$$\frac{dp}{dz} = p \cdot \frac{e^{-z}}{1+e^{-z}}$$

Adding & Subtracting 1 in the numerator

$$= p \cdot \frac{1+e^{-z}-1}{1+e^{-z}}$$

$\frac{1}{1+e^{-z}}$

$$= p \cdot \left(\frac{1+e^{-z}}{1+e^{-z}} - \frac{1}{1+e^{-z}}\right)$$

$$= p \cdot (1-p)$$

Q2. Show that $\frac{dL}{dz} = p - y$

Soln:

$$L = -[y \ln(p) + (1-y)\ln(1-p)]$$

Step 1: Differentiate w.r.t to p.

$$\frac{dL}{dp} = -\left[y\frac{1}{p} + (1-y)\frac{-1}{1-p}\right]$$

Simplifying:

$$\frac{dL}{dp} = \frac{-y}{p} + \frac{1-y}{1-p}$$

From question 1:

$$\frac{dp}{dz} = p(1-p)$$

$$\therefore \quad \frac{dL}{dz} = \frac{dL}{dp} \times \frac{dp}{dz}$$

$$= \left[\frac{-y}{p} + \frac{1-y}{1-p}\right] p(1-p)$$

$$= -y(1-p) + (1-y)p$$

$$= -y + yp + p - yp$$

$$\boxed{\frac{dL}{dz} = p - y}$$

Question 3:- Find $\frac{dL}{dw}$ and $\frac{dL}{db}$

Soln:

$$z = wx + b$$

From Question 2:

$$\frac{dL}{dz} = p - y$$

Step 1:- differentiate z w.r.t w

$$z = wx + b$$

$$\frac{\partial z}{\partial w} = x$$

Step 2: differentiate z w.r.t. b

$$\frac{\partial z}{\partial b} = 1$$

Step 3:- Chain rule

$$\frac{\partial L}{\partial w} = \frac{dL}{dz} \times \frac{\partial z}{\partial w}$$

$$= (p - y) \times x$$

$$\boxed{\frac{\partial L}{\partial w} = (p - y)x}$$

For the bias:

$$\frac{\partial L}{\partial b} = \frac{dL}{dz} \times \frac{\partial z}{\partial b}$$

$$= (p - y) \times 1$$

$$\boxed{\frac{\partial L}{\partial b} = p - y}$$

Question 4: Calculate the Hessian matrix

Sol^{n}: first-order derivatives from Q.3 are:

$$\frac{\partial L}{\partial w} = (p - y)x$$

$$\frac{\partial L}{\partial b} = p - y$$

Step 1: Find the derivative of p with respect to w using the chain rule:

$$\frac{\partial p}{\partial w} = \frac{dp}{dz} \times \frac{\partial z}{\partial w}$$

$$\because \quad \frac{dp}{dz} = p(1 - p)$$

and, $$\frac{\partial z}{\partial w} = x$$

$$\therefore \boxed{\frac{\partial p}{\partial w} = p(1-p)x}$$

Similarly, for bias:

$$\frac{\partial p}{\partial b} = \frac{dp}{dz} \times \frac{\partial z}{\partial b}$$

$$\boxed{\frac{\partial p}{\partial b} = p(1-p)}$$

Step 2: Find the second derivative w.r.t. weight.

$$\frac{\partial L}{\partial w} = (p-y)x$$

$$\frac{\partial^2 L}{\partial w^2} = x \frac{\partial p}{\partial w}$$

$$= x[p(1-p)x]$$

$$\therefore \frac{\partial^2 L}{\partial w^2} = p(1-p)x^2$$

Step 3: Mixed partial derivative:

$$\frac{\partial^2 L}{\partial b \partial w} = \frac{\partial}{\partial b}[(p-y)x]$$

$$= x \frac{\partial p}{\partial b}$$

$$\therefore \boxed{\frac{\partial^2 L}{\partial w \partial b} = p(1-p)x}$$

$$\boxed{\frac{\partial p}{\partial w} = p(1-p)x}$$

Similarly, for bias:

$$\frac{\partial p}{\partial b} = \frac{dp}{dz} \times \frac{\partial z}{\partial b}$$

$$\boxed{\frac{\partial p}{\partial b} = p(1-p)}$$

Step 2: Find the second derivative w.r.t. weight.

$$\frac{\partial L}{\partial w} = (p-y)x$$

$$\frac{\partial^2 L}{\partial w^2} = x \frac{\partial p}{\partial w}$$

$$= x[p(1-p)x]$$

$$\therefore \quad \frac{\partial^2 L}{\partial w^2} = p(1-p)x^2$$

Step 3: Mixed partial derivative:

$$\frac{\partial^2 L}{\partial b \partial w} = \frac{\partial}{\partial b}[(p-y)x]$$

$$= x \frac{\partial p}{\partial b}$$

$$\therefore \quad \boxed{\frac{\partial^2 L}{\partial w \partial b} = p(1-p)x}$$

Step 4: Find the second derivative w.r.t to bias

$$\frac{\partial L}{\partial b} = p - y$$

$$\frac{\partial^2 L}{\partial b^2} = \frac{\partial p}{\partial b}$$

$$\therefore \quad \boxed{\frac{\partial^2 L}{\partial b^2} = p(1-p)}$$

Final Hessian matrix

Putting all three results together:

$$H = \begin{bmatrix} p(1-p)x^2 & p(1-p)x \\ p(1-p)x & p(1-p) \end{bmatrix}$$

Checking answers with the 5 students

At $w = 0$ and $b = 0$:

$$z = wx + b = 0$$

$$\therefore \quad p = \frac{1}{1+e^0} = 0.5$$

$$p(1-p) = 0.05\,(0.5) = 0.25$$

For each student, the hessian becomes:

$$H_i = \begin{bmatrix} 0.25 x_i^2 & 0.25 x_i \\ 0.25 x_i & 0.25 \end{bmatrix}$$



Taking the avg over all 5 students:

$$H = \begin{bmatrix} 0.25\ \text{Average}(x^2) & 0.25\ \text{Average}(x) \\ 0.25\ \text{Average}(x) & 0.25 \end{bmatrix}$$

Your expected result is:

$$H = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.25 \end{bmatrix}$$

Q3. checks are:

$$\boxed{\frac{\partial L}{\partial w} = -0.4}$$

$$\boxed{\frac{\partial L}{\partial b} = 0.1}$$